

Abdoali Mazhar Zakir

(+61) 414 305 580 | abdoali30112004@gmail.com | Sydney, NSW | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Final-year Software Engineering (Honours) student at UNSW who specialises in building scalable data systems and applied machine learning solutions. Interested in digital solutions, automation, and emerging technologies. Experienced working within high-performing, cross-functional teams to deliver data and software solutions, from a financial crime knowledge graph to a self-healing ETL platform. Shipped production-ready applications across the stack, including viticulture weather prediction and AI-driven platforms. Streamlined deployment pipelines to cut manual workflows by 30% and improved system reliability. Currently working on developing machine learning models for complex Citation Link prediction.

EDUCATION

Bachelor of Engineering (Honours) in Software Engineering
University of New South Wales

September 2024 - December 2026

Bachelor of Science in Computer Science & Engineering
Manipal Academy of Higher Education, India

July 2022 - May 2024

EXPERIENCE.

Data Analyst Intern

Futures Sport & Entertainment

July 2026 - September 2026

- Joined the Data Analytics team at Futures Sport & Entertainment, supporting data-driven sponsorship and audience insight work across sport and entertainment clients.
- Working across data collection, cleaning, and analysis to support client-facing reporting and insight generation, three days per week through the internship period.
- Supporting client-facing reporting and communication for sponsorship and audience insight projects, translating data findings into business impact for sport and entertainment clients.

Data Engineer Intern

Tata Consultancy Services (TCS)

January 2026 - February 2026

- Engineered a Financial Crime Compliance knowledge graph by transforming large-scale transaction and account datasets into structured graph entities, delivering a scalable framework for deeper analysis of complex financial networks.
- Applied graph similarity algorithms to detect hidden relationships between accounts, strengthening fraud-risk detection capability by surfacing previously unidentified connections across financial networks.
- Presented technical data findings and graph-based methodology to industry stakeholders, translating complex metrics into plain-language risk insights that improved stakeholder decision-making.
- Collaborated within a multi-disciplinary Agile engineering team on project planning and milestone reviews, contributing to the successful delivery of final technical presentations to industry stakeholders.

PROJECTS

Sentinel ETL Self-Healing Data Platform Personal Project

June 2026

- Independently architected an enterprise-inspired ETL self-healing data quality engine (Python/FastAPI) that automatically detects and remediates data inconsistencies. Demonstrating curiosity and initiative in exploring emerging data engineering technologies outside of coursework.
- Engineered a platform implementing Medallion Architecture (Bronze/Silver/Gold layers) to ingest, validate, repair, and transform datasets automatically.
- Developed a Next.js/TypeScript/Tailwind analytics dashboard to visualise data quality scores, pipeline health, and dataset history. Used DuckDB, Pandas, and PyArrow for the data processing layer, PostgreSQL for metadata storage. [GitHub: [Sentinel-ETL](#)]

TerraCast: Viticulture Weather & Advisory Platform UNSW

February 2026 – May 2026

- Automated data ingestion and cleansing workflows on AWS Lambda, unifying disparate weather variables rainfall, frost, temperature, and wind into a standardised pipeline for downstream forecasting models.
- Architected responsive frontend features using Next.js (App Router), including dashboards and recommendation views, with state management and API integration. Supporting sustainable, data-driven decision-making for viticulture resource management.
- Implemented server-side generative AI features using OpenAI API (gpt-4o-mini), including a viticulture assistant and forecast-based recommendation system using structured outputs.
- Collaborated in an Agile team, contributing to API design, system integration, and technical documentation, including testing strategy and deployment planning.

EXTRACURRICULAR ACTIVITIES

Software Developer Enigma Technical Society

September 2023 – May 2024

- Built internal reporting and automation scripts in Python to track and analyse society operations data, cutting manual reporting effort by 30%.
- Designed and maintained a PostgreSQL data layer for event and membership data, ensuring clean, query-ready datasets for a 6-member Agile team.
- Automated data-refresh and deployment workflows using GitHub Actions CI/CD pipelines, reducing manual errors in recurring data updates.

CORE COMPETENCIES

Languages: Python, JavaScript, C++, SQL, Cypher, HTML/CSS.

Frameworks & Libraries: Node.js, React, Jest, Pytest, PyTorch

Databases: PostgreSQL, MySQL, Neo4j(graph)

Tools & Platforms: Docker, AWS (Terraform, S3), Databricks, Postman, Neo4j

VOLUNTEERING AND SOCIETIES

Lead Volunteer Voluntary Service Overseas (VSO), Manipal

June 2023- May 2024

Secretary Social Football Society, UNSW

March 2024- Present

Sports Captain International Centre for Applied Science, Manipal University

June 2023- May 2024

